

Nuclear-Norm Best Approximation at Rank-Deficient Matrix-Line Crossings: Certificate Geometry and Singular Smoothing

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S1 Index and scope of the supplement

This supplement contains auxiliary results that are useful for implementing or embedding the inner oracle but are not needed for the main theorem chain. The essential proofs of compact-root failure, certificate completion, Fenchel recovery/selection, and the singular rates remain in the main paper.

1. **Exact interval bisection.** We record the elementary convergence proof for the set-valued scalar update used by the exact reference oracle.
2. **Spectral-level-set normal geometry.** We derive the simple-top normal, distinguish the repeated-top Bouligand tangent cone, quantify radial drift, and give exact projective normal-error formulas. This material justifies the rank-one normal in the motivating spectral-level-set application; the main matrix-line results permit general normals.
3. **Feasible primal–dual certificate.** We give an exact-real repair and a gap decomposition for an approximate smooth direction, while separating those identities from the stronger requirements of a machine-validated implementation.
4. **Detailed reproducibility tables.** We report the full constant-level checks for the regular, strict, boundary, moving-crossover, smoothing-family, direction, and higher-order cancellation formulas summarized in the main paper.

S2 Exact interval bisection

Let $L(\lambda) = a(\lambda) - \beta(\lambda)$ and $U(\lambda) = a(\lambda) + \beta(\lambda)$, with the notation of Theorem 2.2.

Proposition S2.1 (Set-valued bisection). *If $U(\lambda) < 0$, every minimizer lies to the right of λ ; if $L(\lambda) > 0$, every minimizer lies to the left; and if $L(\lambda) \leq 0 \leq U(\lambda)$, then λ is a minimizer. Starting from the bracket in (8), midpoint updates either stop at an exact minimizer or generate nested intervals converging to one.*

Proof. The first two cases say that the entire scalar subdifferential is strictly negative or strictly positive, so one-dimensional convexity identifies the side containing every minimizer. The third case is $0 \in \partial\phi(\lambda)$. Each update preserves a minimizer, and nested closed brackets whose lengths tend to zero converge to a point of the closed minimizer set. $\square$

S3 The Normal Θ : Origin, Geometry, and Computability

This section supplies the geometry of the motivating spectral-level-set application. It answers four distinct questions: what motivates a spectral scale, which extra modeling choice creates an exact level set, why its normal is $\Theta = u_1 v_1^\top$, and what can actually be certified when the leading singular vectors are computed approximately.

S3.1 Spectral scaling is the motivation, not the equality constraint

Let $W \in \mathbb{R}^{m \times n}$ represent a hidden linear map $y = Wx$. Since

$$\frac{\|Wx\|_2/\sqrt{m}}{\|x\|_2/\sqrt{n}} \leq \|W\|_2 \sqrt{\frac{n}{m}},$$

the spectral formulation of width-stable feature learning motivates the scale [3]

$$\|W\|_2, \|\Delta W\|_2 = \Theta\left(\sqrt{\frac{m}{n}}\right) \quad (\text{N1})$$

Equation (N1) is an asymptotic order statement. It does *not* require the exact equality $\|W\|_2 = R$ at every iteration. The asymptotic notation $\Theta(\cdot)$ in (N1) is unrelated to the matrix normal $\Theta(W)$ defined below; the shared glyph should not be read as a shared mathematical object. Following Xie et al. [2], we make the additional hard-constraint modeling choice

$$\mathcal{L}_R = \{W \in \mathbb{R}^{m \times n} : \|W\|_2 = R\}, \quad R \asymp \sqrt{m/n}. \quad (\text{N2})$$

The normal Θ is a consequence of this additional level-set model, not an independent optimizer ingredient and not a consequence of μP alone.

S3.2 The smooth stratum and its unit normal

Write the singular values of W as $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_q$, where $q = \min\{m, n\}$. Throughout this subsection $R > 0$. Statements below that invoke σ_2 , a second singular subspace, or a two-leading-value eigensolver assume $q \geq 2$. The one-dimensional case is elementary and is excluded from those statements rather than hidden behind an undefined σ_2 .

Lemma S3.1 (Spectral-norm differential and generalized normal). *If $\sigma_1(W) > \sigma_2(W)$, then the spectral norm is real analytic in a neighborhood of W , and*

$$D\|W\|_2[E] = \langle u_1 v_1^\top, E \rangle. \quad (\text{N3})$$

Consequently

$$\Theta(W) := u_1(W) v_1(W)^\top \quad (\text{N4})$$

is the unique outward, gradient-oriented Frobenius unit normal to the smooth stratum of $\mathcal{L}_R$; the unoriented normal space is $\text{span}\{\Theta(W)\}$. It is invariant under the joint sign change $(u_1, v_1) \mapsto (-u_1, -v_1)$, has rank one, and satisfies

$$\|\Theta\|_2 = \|\Theta\|_F = \|\Theta\|_* = 1. \quad (\text{N5})$$

If the positive leading singular value has multiplicity $k > 1$, then the spectral norm is not differentiable and

$$\partial\|W\|_2 = \{U_k S V_k^\top : S \succeq 0, \text{tr } S = 1\}. \quad (\text{N6})$$

Proof. Apply simple-eigenvalue perturbation theory to the symmetric dilation

$$\mathcal{H}(W) = \begin{pmatrix} 0 & W \\ W^\top & 0 \end{pmatrix}.$$

Its largest eigenvalue is $\sigma_1(W)$, with normalized eigenvector $(u_1^\top, v_1^\top)^\top / \sqrt{2}$. Differentiation gives (N3). Simplicity makes the one-dimensional left and right singular spaces unique up to a joint sign, which cancels in the outer product. The norm identities follow because Θ has one nonzero singular value equal to one. Formula (N6) is the standard spectral-norm subdifferential formula [1]. $\square$

Proposition S3.2 (Bouligand tangent cone at a repeated leading singular value). *Let $W \in \mathcal{L}_R$ have a positive leading singular value R of multiplicity k , with corresponding matrices U_k, V_k . Then the Bouligand tangent cone of the equality level set is*

$$T_{\mathcal{L}_R}^B(W) = \{E : \lambda_{\max}(\text{sym}(U_k^\top E V_k)) = 0\}, \quad \text{sym}(B) = \frac{B + B^\top}{2}. \quad (\text{N6a})$$

For $k = 1$, this reduces to the hyperplane $\{E : \langle u_1 v_1^\top, E \rangle_F = 0\}$. For $k > 1$, it is generally a nonlinear, nonconvex cone and cannot be obtained by choosing one convenient matrix from $\partial\|W\|_2$.

Proof. The directional derivative of the largest singular value at a multiplicity k cluster is

$$\|\cdot\|_2'(W; E) = \lambda_{\max}(\text{sym}(U_k^\top EV_k)). \quad (\text{N6b})$$

This follows by compressing the perturbation of the symmetric dilation $\mathcal{H}(W)$ to its leading eigenspace. For necessity, take any $E \in T_{\mathcal{L}_R}^B(W)$. By definition there are $t_j \downarrow 0$ and $E_j \rightarrow E$ such that $W + t_j E_j \in \mathcal{L}_R$. The spectral norm is Hadamard directionally differentiable, hence

$$0 = \lim_{j \rightarrow \infty} \frac{\|W + t_j E_j\|_2 - \|W\|_2}{t_j} = \|\cdot\|_2'(W; E).$$

For sufficiency, suppose the derivative in (N6b) is zero and define

$$W(t) = R \frac{W + tE}{\|W + tE\|_2}.$$

Then $W(t) \in \mathcal{L}_R$ and $\|W + tE\|_2 = R + o(t)$, so $W(t) = W + tE + o(t)$. Thus E is a Bouligand tangent direction. $\square$

Boundary of the claim. The dual/weight symmetry gap. The manuscript treats rank deficiency of the *dual matrix* $G + \lambda\Theta$ set-valuedly because that inclusion is the tractable inner problem under study. At a repeated leading singular value of the *weight*, it does not select one element of $\partial\|W\|_2$ and call the resulting hyperplane the tangent space. Proposition S3.2 shows the actual first-order object. The present inner-oracle analysis instead excludes this stratum from any outer claim. That is a genuine asymmetry of scope, not a resolution of the global stratified problem; a tangent-cone/normal-cone outer method remains open.

Remark S3.3 (A normal is not a projector). The matrix Θ is a unit normal, not a projector. Under the Frobenius metric the orthogonal tangent projector is the linear map

$$\Pi_{T_W}(E) = E - \langle \Theta, E \rangle \Theta.$$

Lemma S3.4 (Dimension-free sensitivity of the normal). *Let $\gamma_W = \sigma_1(W) - \sigma_2(W) > 0$. For every perturbation E ,*

$$\|D\Theta(W)[E]\|_F \leq \frac{\sqrt{2}}{\gamma_W} \|E\|_2. \quad (\text{N7})$$

Proof. Choose differentiable singular-vector representatives satisfying $u_1^\top Du_1 = v_1^\top Dv_1 = 0$. The simple-eigenvector perturbation formula for $\mathcal{H}(W)$ gives

$$\left\| D \left(\frac{u_1}{\sqrt{2}}, \frac{v_1}{\sqrt{2}} \right) [E] \right\|_2 \leq \frac{\|E\|_2}{\gamma_W}.$$

Since

$$D\Theta[E] = (Du_1[E])v_1^\top + u_1(Dv_1[E])^\top$$

and the two summands are Frobenius orthogonal,

$$\|D\Theta[E]\|_F^2 = \|Du_1[E]\|_2^2 + \|Dv_1[E]\|_2^2,$$

which proves (N7). $\square$

S3.3 Tangent constraint, ball oracle, and the role of retraction

On the simple-leading-value stratum, the tangent space is the kernel of (N3):

$$T_W \mathcal{L}_R = \{\Phi : \langle \Theta(W), \Phi \rangle = 0\}. \quad (\text{N8})$$

For a step $E = -\eta R\Phi$, this also follows from

$$\|W - \eta R\Phi\|_2 = R - \eta R \langle \Theta, \Phi \rangle + O\left(\frac{\eta^2 R^2}{\gamma_W}\right). \quad (\text{N9})$$

Thus the tangent LMO studied below is

$$p^* = \max_{\|\Phi\|_2 \leq 1, \langle \Theta, \Phi \rangle = 0} \langle G, \Phi \rangle. \quad (\text{P})$$

Here Θ is computed from the current weight W , whereas G is a loss-gradient or momentum-like matrix supplied to the oracle. They are different inputs. Their only coupling in the exact dual is through the affine line $G + \lambda\Theta$.

Lemma S3.5 (Closed ball versus unit sphere). *If $p^* > 0$, every optimizer of (P) saturates $\|\Phi\|_2 = 1$; hence the ball and sphere formulations have the same optimal value and the same optimizers. If $p^* = 0$, the optimal values still agree provided $mn \geq 2$, but the ball formulation also contains interior optima. When $mn = 1$, the tangent hyperplane is $\{0\}$, so the unit-sphere problem has no feasible point.*

Proof. If an optimizer with positive objective had spectral norm strictly below one, positive rescaling would remain tangent and feasible while increasing the objective. If $p^* = 0$, central symmetry of the feasible set implies that $\langle G, \Phi \rangle = 0$ for every tangent feasible Φ : otherwise either Φ or $-\Phi$ would have positive objective. When $mn \geq 2$, the nonzero normal Θ has a nonzero Frobenius-orthogonal complement. Choose $E \neq 0$ in that complement and replace it by $E/\|E\|_2$. This gives a tangent matrix of unit spectral norm, proving equality of the optimal values. The $mn = 1$ exception is immediate. $\square$

This lemma explains why convex duality must be established on the closed ball, even if an implementation later normalizes a nonzero accepted direction to unit spectral norm.

Lemma S3.6 (Explicit second-order drift). *Let $\gamma_W > 0$ and $\|E\|_2 \leq \gamma_W/4$. Then*

$$0 \leq \|W + E\|_2 - \|W\|_2 - \langle \Theta, E \rangle \leq \frac{2\|E\|_2^2}{\gamma_W}. \quad (\text{N10})$$

In particular, a tangent step $E = -\eta R\Phi$ with $\|\Phi\|_2 \leq 1$ and $\eta R \leq \gamma_W/4$ has radial drift at most

$$\frac{2\eta^2 R^2}{\gamma_W}. \quad (\text{N11})$$

Proof. Convexity gives the lower bound. In a basis whose first vector is the leading eigenvector of $\mathcal{H}(W)$, write the symmetric dilation and its perturbation as

$$\mathcal{H}(W) = \begin{pmatrix} \sigma_1 & 0 \\ 0 & B \end{pmatrix}, \quad \mathcal{H}(E) = \begin{pmatrix} a & b^\top \\ b & C \end{pmatrix},$$

with $\lambda_{\max}(B) \leq \sigma_1 - \gamma_W$ and $\rho := \|E\|_2 = \|\mathcal{H}(E)\|_2$. The Schur-complement equation for the perturbed leading eigenvalue gives

$$0 \leq \tilde{\sigma}_1 - \sigma_1 - a \leq \frac{\rho^2}{\gamma_W - 2\rho} \leq \frac{2\rho^2}{\gamma_W}.$$

Here $a = \langle \Theta, E \rangle$. The dilation may have additional zero eigenvalues when $m \neq n$; their distance from σ_1 is at least $\sigma_1 \geq \gamma_W$, so the same bound applies. $\square$

For $W + E \neq 0$, define the radial retraction

$$\text{Retr}_W(E) = R \frac{W + E}{\|W + E\|_2}. \quad (\text{N12})$$

Its derivative is

$$D \text{Retr}_W(0)[E] = E - \frac{W}{R} \langle \Theta, E \rangle. \quad (\text{N13})$$

Thus retraction guarantees exact radial feasibility, whereas the tangent LMO chooses the best first-order direction under a prescribed spectral budget. They are not interchangeable. Retraction also does not automatically preserve a fixed positive leading spectral gap.

S3.4 Approximate normals live in a projective geometry

An implementation computes an approximate singular triplet $(\hat{\sigma}, \hat{u}, \hat{v})$, typically by a few power or block iterations, and sets

$$\hat{\Theta} = \hat{u}\hat{v}^\top. \quad (\text{N14})$$

For classical bilateral power iteration with a starting vector having nonzero leading-component overlap, the right-vector angle obeys

$$\tan \angle(\hat{v}_k, v_1) \leq \left(\frac{\sigma_2(W)}{\sigma_1(W)} \right)^{2k} \tan \angle(\hat{v}_0, v_1). \quad (\text{N14a})$$

Thus a fixed iteration budget is reliable only when the leading ratio is separated from one.

For a finite iteration budget, (N14a) applies with the actual initial angle; a warm start changes that initial angle, not the exponent. The equality $\langle \hat{\Theta}, \Phi \rangle = 0$ is unchanged under $\hat{\Theta} \mapsto -\hat{\Theta}$. The relevant object is therefore the normal *line*, not an oriented normal matrix.

Let

$$\alpha_u = \arccos |u_1^\top \hat{u}|, \quad \alpha_v = \arccos |v_1^\top \hat{v}|, \quad \alpha_u, \alpha_v \in [0, \pi/2], \quad (\text{N15})$$

and put $\vartheta = \alpha_u + \alpha_v$.

Lemma S3.7 (Exact projective distance between rank-one normals).

$$\min_{s \in \{\pm 1\}} \left\| \Theta - s\hat{\Theta} \right\|_* = 2 \sin \frac{\vartheta}{2}, \quad \min_{s \in \{\pm 1\}} \left\| \Theta - s\hat{\Theta} \right\|_F = \sqrt{2 - 2 \cos \alpha_u \cos \alpha_v}. \quad (\text{N16})$$

Proof. Choose s and a factorization $s\hat{\Theta} = \tilde{u}\tilde{v}^\top$ so that $u_1^\top \tilde{u} = \cos \alpha_u \geq 0$ and $v_1^\top \tilde{v} = \cos \alpha_v \geq 0$. The difference has rank at most two. The sum and product of its two squared singular values are

$$2 - 2 \cos \alpha_u \cos \alpha_v, \quad \sin^2 \alpha_u \sin^2 \alpha_v.$$

Consequently the squared nuclear norm is

$$2 - 2 \cos \alpha_u \cos \alpha_v + 2 \sin \alpha_u \sin \alpha_v = 4 \sin^2 \frac{\alpha_u + \alpha_v}{2}.$$

The other sign gives the larger of the two projective representatives. $\square$

Theorem S3.8 (Sharp worst-case leakage of an estimated tangent space). *For unit rank-one normals $\Theta, \hat{\Theta}$,*

$$\sup_{\substack{\|\Phi\|_2 \leq 1 \\ \langle \hat{\Theta}, \Phi \rangle = 0}} |\langle \Theta, \Phi \rangle| = \begin{cases} \sin \vartheta, & 0 \leq \vartheta \leq \pi/2, \\ 1, & \pi/2 \leq \vartheta \leq \pi. \end{cases} \quad (\text{N17})$$

Proof. The absolute value permits either sign of the objective. After the projective orientation used in the proof of lemma S3.7, apply T1 strong duality (Theorem 2.1) to the oracle with objective Θ and normal $\hat{\Theta}$:

$$\min_{\lambda \in \mathbb{R}} \left\| \Theta + \lambda \hat{\Theta} \right\|_*.$$

For $\lambda = -t$, $t \geq 0$, the squared nuclear norm is

$$1 + t^2 - 2t \cos(\alpha_u + \alpha_v).$$

Its minimum is $\sin^2 \vartheta$ when $\cos \vartheta \geq 0$, and 1 otherwise. The branch $\lambda = t \geq 0$ has squared norm

$$1 + t^2 + 2t \cos(\alpha_u - \alpha_v),$$

whose minimum is 1 because $|\alpha_u - \alpha_v| \leq \pi/2$. Comparing the branches proves (N17). $\square$

Boundary of the claim. Theorem S3.8 is a sharp *uniform worst-case upper bound*. It is not a lower bound on every realized oracle direction: a particular direction may be tangent to both normal hyperplanes. Likewise, an angle measured directly between two normal matrices is not, in general, the sum $\alpha_u + \alpha_v$ appearing in (N17). No empirical leakage percentage is inferred in this paper because the required pair (α_u, α_v) was not recorded.

S4 A Feasible Certificate and Its Finite-Precision Boundary

An approximate matrix output cannot be certified merely by reporting a polar residual and a tangent residual. If the tangent residual is nonzero, the matrix is not feasible for the primal oracle, so its objective value need not be a valid primal lower bound.

Assume here that $\Theta = uv^\top$ is a unit rank-one normal. Given an arbitrary approximate matrix Z , define the spectral normalization

$$\bar{Z} = \frac{Z}{\max\{1, \|Z\|_2\}}, \quad c = \langle \Theta, \bar{Z} \rangle, \quad (54)$$

and the feasible repair

$$\tilde{Z} = \frac{\bar{Z} - c\Theta}{1 + |c|}. \quad (55)$$

Theorem S4.1 (Feasible repair and valid primal–dual gap in exact arithmetic). *The repaired matrix satisfies*

$$\langle \Theta, \tilde{Z} \rangle = 0, \quad \|\tilde{Z}\|_2 \leq 1. \quad (56)$$

For any multiplier λ , define

$$\text{Gap}(\lambda, Z) = \|G + \lambda\Theta\|_* - \langle G, \tilde{Z} \rangle. \quad (57)$$

Then

$$\text{Gap}(\lambda, Z) \geq p^* - \langle G, \tilde{Z} \rangle \geq 0. \quad (58)$$

Thus (57) is a valid upper bound on the primal LMO suboptimality.

If

$$r_{\text{pol}} = \|G + \lambda\Theta\|_* - \langle G + \lambda\Theta, \bar{Z} \rangle, \quad (59)$$

then $r_{\text{pol}} \geq 0$, and

$$\text{Gap}(\lambda, Z) \leq r_{\text{pol}} + |\lambda||c| + \frac{2\|G\|_*|c|}{1 + |c|}. \quad (60)$$

Finally, if a computable quantity $\hat{\phi}(\lambda)$ has been proved to satisfy

$$\hat{\phi}(\lambda) \geq \|G + \lambda\Theta\|_*,$$

then

$$\hat{\phi}(\lambda) - \langle G, \tilde{Z} \rangle \quad (61)$$

is also a valid certificate.

Proof idea. First normalize into the spectral ball. Then subtract the normal component and divide by a worst-case triangle-inequality factor so that tangency and spectral feasibility both hold exactly. Only after this repair is the dual-minus-primal difference a legitimate gap.

What this result does. The repair turns any approximate polar map—including one affected by truncation, finite precision, or an imperfect multiplier—into a direction that the inner oracle may accept at a prescribed objective accuracy. The certificate measures the price of the repair rather than silently treating an infeasible matrix as primal feasible.

Proof. By definition, $\|\bar{Z}\|_2 \leq 1$. Because the rank-one unit normal satisfies $\langle \Theta, \Theta \rangle = \|\Theta\|_{\text{F}}^2 = 1$,

$$\langle \Theta, \tilde{Z} \rangle = \frac{c - c}{1 + |c|} = 0.$$

Furthermore,

$$\|\bar{Z} - c\Theta\|_2 \leq \|\bar{Z}\|_2 + |c|\|\Theta\|_2 \leq 1 + |c|.$$

Dividing by $1 + |c|$ proves (56).

Weak duality from Theorem 2.1 gives

$$\|G + \lambda\Theta\|_* \geq p^*.$$

Since $\tilde{Z}$ is primal feasible,

$$\langle G, \tilde{Z} \rangle \leq p^*.$$

Subtracting proves (58).

For the decomposition, put $A = G + \lambda\Theta$. Adding and subtracting $\langle A, \bar{Z} \rangle$ gives

$$\begin{aligned} \text{Gap}(\lambda, Z) &= \|A\|_* - \langle G, \tilde{Z} \rangle \\ &= r_{\text{pol}} + \lambda c + \langle G, \bar{Z} - \tilde{Z} \rangle. \end{aligned}$$

Equation (55) gives

$$\bar{Z} - \tilde{Z} = \frac{|c|\bar{Z} + c\Theta}{1 + |c|}.$$

By spectral/nuclear duality,

$$\begin{aligned} \left| \langle G, \bar{Z} - \tilde{Z} \rangle \right| &\leq \frac{|c| \|G\|_* \|\bar{Z}\|_2 + |c| \|G\|_* \|\Theta\|_2}{1 + |c|} \\ &\leq \frac{2 \|G\|_* |c|}{1 + |c|}. \end{aligned}$$

Using $\lambda c \leq |\lambda||c|$ proves (60). Replacing the exact dual value by any proved upper bound preserves the coverage inequality, giving (61). $\square$

Remark S4.2 (The same repair for an arbitrary nonzero normal). The rank-one normalization makes the formulas particularly simple but is not essential. For any nonzero Θ , normalize Z as in (54) and put

$$b = \langle \Theta, \bar{Z} \rangle, \quad d = \frac{b}{\|\Theta\|_F^2}, \quad \eta = |d| \|\Theta\|_2, \quad \tilde{Z} = \frac{\bar{Z} - d\Theta}{1 + \eta}.$$

Then $\langle \Theta, \tilde{Z} \rangle = 0$ and $\|\tilde{Z}\|_2 \leq 1$. With $r_{\text{pol}} = \|G + \lambda\Theta\|_* - \langle G + \lambda\Theta, \bar{Z} \rangle$, the same calculation gives

$$0 \leq p^* - \langle G, \tilde{Z} \rangle \leq \|G + \lambda\Theta\|_* - \langle G, \tilde{Z} \rangle \leq r_{\text{pol}} + |\lambda| |b| + \frac{2 \|G\|_* \eta}{1 + \eta}.$$

Indeed, $\bar{Z} - \tilde{Z} = (\eta\bar{Z} + d\Theta)/(1 + \eta)$, whose spectral norm is at most $2\eta/(1 + \eta)$. This is an elementary extension of the same feasibility argument, not a further smoothing law.

Boundary of the claim. The feasibility and coverage statements above are exact-real-arithmetic identities. A machine-rigorous certificate additionally needs outward-rounded upper bounds for the nuclear norm and spectral normalization, a downward- rounded primal objective, and enclosures for the tangency calculation. Plain double-precision evaluation is a numerical check, not by itself a validated certificate. These are necessary ingredients, not a complete validated implementation: an interval enclosure of the tangency residual alone does not certify exact membership in the hyperplane unless the final repair is itself performed exactly or with an interval-safe correction.

Corollary S4.3 (Optional unit-spectral-norm output). *Let*

$$P = \bar{Z} - c\Theta.$$

If $P \neq 0$, choose

$$s_G = \begin{cases} 1, & \langle G, P \rangle \geq 0, \\ -1, & \langle G, P \rangle < 0, \end{cases} \quad Z_{\text{unit}} = s_G \frac{P}{\|P\|_2}. \quad (62)$$

Then

$$\langle \Theta, Z_{\text{unit}} \rangle = 0, \quad \|Z_{\text{unit}}\|_2 = 1,$$

and

$$\langle G, Z_{\text{unit}} \rangle \geq \langle G, \tilde{Z} \rangle. \quad (63)$$

Consequently, replacing $\tilde{Z}$ by Z_{unit} never worsens the valid primal–dual certificate.

Likewise, if $p^* > 0$, then the smooth solution $\Phi_\delta \neq 0$ may be replaced by

$$\Phi_\delta^{\text{unit}} = \frac{\Phi_\delta}{\|\Phi_\delta\|_2}.$$

It remains tangent, has unit spectral norm, and has no smaller original linear objective. The lower bound

$$\sigma_1(G + \lambda_\delta \Theta) \geq s_0 := p^*/q > 0, \quad q = \min\{m, n\},$$

holds uniformly for every $\delta > 0$, because $q\sigma_1(G + \lambda\Theta) \geq \|G + \lambda\Theta\|_* \geq p^*$ for every λ . Therefore the rescaling factor satisfies

$$\frac{1}{\|\Phi_\delta\|_2} = \sqrt{1 + \frac{\delta^2}{\sigma_1(G + \lambda_\delta \Theta)^2}} = 1 + O(\delta^2). \quad (64)$$

Proof. Tangency and unit norm in (62) are immediate. Moreover, $\|P\|_2 \leq 1 + |c|$, so if $\langle G, P \rangle \geq 0$,

$$\langle G, Z_{\text{unit}} \rangle = \frac{\langle G, P \rangle}{\|P\|_2} \geq \frac{\langle G, P \rangle}{1 + |c|} = \langle G, \tilde{Z} \rangle.$$

If $\langle G, P \rangle < 0$, the left side is positive whereas the right side is negative.

For the smooth direction, comparison with the feasible zero direction in (34) gives

$$\langle G, \Phi_\delta \rangle \geq \delta R(\Phi_\delta) \geq 0.$$

To see rigorously that $\Phi_\delta \neq 0$ when $p^* > 0$, let Φ^* be an exact primal optimizer. For sufficiently small $t > 0$, the exact identity $1 - \sqrt{1 - z^2} = z^2/(1 + \sqrt{1 - z^2}) \leq z^2$ gives

$$R(t\Phi^*) \leq t^2 \|\Phi^*\|_F^2.$$

Consequently

$$\langle G, t\Phi^* \rangle - \delta R(t\Phi^*) \geq tp^* - \delta t^2 \|\Phi^*\|_F^2 > 0,$$

whereas the regularized objective at zero is zero. Hence the unique smooth optimizer is nonzero, and positive radial rescaling cannot decrease its linear objective. Finally, (29) gives

$$\|\Phi_\delta\|_2 = \frac{\sigma_1(G + \lambda_\delta \Theta)}{\sqrt{\sigma_1(G + \lambda_\delta \Theta)^2 + \delta^2}},$$

which proves (64). □

Boundary of the claim. The unit repair is certified primal post-processing and an experimental scale-control device. The map $P \mapsto P/\|P\|_2$ is discontinuous at $P = 0$ and may amplify numerical noise nearby. Moreover, $\Phi_\delta^{\text{unit}}$ is no longer the exact optimizer of the regularized primal at finite δ , so its Fenchel center-path identity must not be asserted pointwise. For a fixed instance with $p^* > 0$, however, its rescaling is $1 + O(\delta^2)$ and it has the same limiting center.

S4.1 How to read the decomposition

The three terms in (60) have different meanings.

1. r_{pol} measures matrix-function or polar optimality error.
2. $|\lambda||c|$ measures the coupling between multiplier size and tangent residual.
3. The last term is the explicit price of moving the infeasible output into the primal feasible set.

Certified stopping rule. Given a target inner error budget ε_t , stop the inner oracle only when a valid quantity of the form (57) or (61) is at most ε_t . Reporting only $|c|$ is not sufficient.

The matrix Z in these diagnostics is the actual returned output of the implemented map. Replacing it by $S_c(G + \lambda\Theta)$, or computing the tangent residual with an estimated normal while interpreting it as a residual for the true normal, changes the quantity being assessed. A small tangent residual alone also does not imply a small r_{pol} or a small feasible gap. Conversely, a small computed floating-point gap is evidence for numerical accuracy only within its recorded norm and pairing errors; it is not a machine-rigorous certificate without validated bounds.

S4.2 A certifiable reference homotopy oracle

The theory suggests the following implementation pattern.

1. **Choose a smoothing level.** Start from a moderate δ ; warm-start it from the previous oracle call when appropriate.
2. **Bracket the smooth multiplier.** Use the proved multiplier bound (47), or a tighter bracket whose coverage has been verified.
3. **Solve the continuous root.** Apply bisection or a safeguarded scalar solver to $h_\delta(\lambda) = 0$. Unlike the compact selection, this function is continuous and strictly increasing.
4. **Form the smooth direction.** Evaluate $Z_\delta(G + \lambda\Theta)$. If the scalar root is only approximate or the matrix function is computed in finite precision, apply (54)–(55).
5. **Certify.** Compute (57), or a proved upper version (61).
6. **Refine only when needed.** If the certificate exceeds the requested tolerance ε_t , reduce δ , tighten the scalar root, or switch to the exact set-valued SVD oracle. Otherwise accept the repaired direction.

This homotopy is “accuracy on demand” only after the normal accuracy and any assumptions required by a downstream method have been certified independently. Theorem S4.1 makes only the *inner* oracle gap computable; it does not certify the normal or an outer descent model. Consequently this is a certifiable reference workflow, not a claim of a fully deployable per-step large-model routine.

Remark S4.4 (Conservatism). The repair is deliberately simple and globally safe. It may be conservative: projecting onto the exact intersection of a spectral ball and a hyperplane could produce a tighter primal value, but would itself require a nontrivial optimization. The displayed repair prioritizes a closed-form guarantee.

S5 Detailed numerical verification

The local laws of Section 5 predict constants, not merely exponents. The file `verify_publication_tables.py` generates every numerical table and the displayed decimal constants in this section from one machine-readable snapshot. It uses 80-digit mpmath arithmetic (100-digit Decimal arithmetic for the moving family), and repeats the non-diagonal calculation at 100 digits. The repeated ratios agree within 10^{-60} and its intrinsic constants within 10^{-65} ; we report fewer digits. These are working precisions and numerical cross-checks, not validated error bounds. Bisection and explicit spectral formulas avoid float64 subtraction of nearby multipliers or directions. The full suite also retains independent float64 regression tests; no assertion is silently disabled under `python -O`. The following commands run the suite and compare the stored JSON/LaTeX snapshots, respectively:

```
python verify_all.py
python verify_publication_tables.py -check
```

Omit `-check` from the second command to regenerate the snapshots.

For the first group, $\Theta = e_1 e_1^\top$ and $G = \begin{pmatrix} 0 & b_1 \\ b_2 & c \end{pmatrix}$.

For the regular row, $b_1 = b_2 = 1$, $c = 1/2$, and $\lambda^* = 1/2$; the ratio is $(\lambda_\delta - \lambda^*)/\delta^2$. For the strict row, $b_1 = b_2 = 1/2$, $c = 1$, and $\lambda^* = 1/4$; the ratio is $(\lambda_\delta - \lambda^*)/\delta$. For the boundary row, $b_1 = b_2 = c = 1$, $\lambda^* = 1$, and the ratio is $(\lambda^* - \lambda_\delta)/\delta^{2/3}$.

Regime	δ	Normalized displacement	Limit
regular	10^{-4}	-1.77777776645	-1.7777777778
regular	10^{-5}	-1.77777777766	-1.7777777778
strict	10^{-6}	-0.3227484177	-0.3227486122
strict	10^{-8}	-0.3227486102	-0.3227486122
boundary	10^{-9}	1.2599202562	1.2599210499
boundary	10^{-12}	1.2599210420	1.2599210499

For the same boundary data, standard Huber gives $(\lambda^* - \lambda_\delta^{\text{hub}})/\delta = 1.9999999500$ at $\delta = 10^{-8}$, versus $1/\beta = 2$. The scaled-Huber regression in the code also confirms that moving the first saturation threshold to $L = 1/2$ changes the constant to $L/\beta = 1$, as required by (84).

For the admissible profile $\psi'(x) = x/(1+x^3)^{1/3}$, $x \geq 0$, we divide the displacement by $\delta^{3/4}$; for Huber and scaled Huber we divide it by δ . Thus the complete profile table tests the tail-dependent exponent as well as the leading constant.

Profile	δ	Normalized displacement	Limit
Huber	10^{-4}	1.999500189915	2.000000000000
Huber	10^{-6}	1.999995000019	2.000000000000
Huber	10^{-8}	1.999999950000	2.000000000000
$p = 3$	10^{-3}	1.272747989329	1.277886208493
$p = 3$	10^{-6}	1.277857161983	1.277886208493
$p = 3$	10^{-9}	1.277886045147	1.277886208493
$p = 3$	10^{-12}	1.277886207574	1.277886208493
$L = 1/2$	10^{-8}	0.999999987500	1.000000000000

The preceding instances are 2×2 . To exercise the general completion and arbitrary-corank strict law, take

$$A^* = \text{diag}(1, 1, 0, 0), \quad \Theta = \text{diag}\left(\frac{3}{8}, \frac{3}{8}, 1, \frac{1}{2}\right), \quad \lambda^* = 1, \quad G = A^* - \Theta.$$

Then $a = 3/4$, $C = \text{diag}(1, 1/2)$, and $\beta = 3/2$. Hard water-filling gives $\eta_F = 3/5$, $K_F = -\text{diag}(3/5, 3/10)$, and $\|K_F\|_F = \sqrt{0.45}$. The smooth-center equation gives

$$\eta_R = 0.715080068980, \quad K_R = -\text{diag}(0.581665969555, 0.336668060890),$$

with $\|K_F - K_R\|_F = 0.040996138377$. Writing Φ_F^* and Φ_R^* for the two corresponding completed certificates, the actual smoothing path gives

δ	$\lambda_\delta - \lambda^*$	$\ \Phi_\delta - \Phi_R^*\ _F$	$\ \Phi_\delta - \Phi_F^*\ _F$
10^{-2}	-7.150296×10^{-3}	7.896227×10^{-5}	4.098952×10^{-2}
10^{-3}	-7.150796×10^{-4}	7.858742×10^{-7}	4.099607×10^{-2}
10^{-4}	-7.150801×10^{-5}	7.854955×10^{-9}	4.099614×10^{-2}
10^{-5}	-7.150801×10^{-6}	7.854576×10^{-11}	4.099614×10^{-2}

Thus the path converges to the smooth rather than the minimum-Frobenius center, while $(\lambda_\delta - \lambda^*)/\delta \rightarrow -\eta_R$, as predicted by Theorems 4.3 and 5.3 and Corollary 4.4. On the same line, the standard Huber path gives slope $-\eta_F = -3/5$ and converges to Φ_F^* to below 10^{-11} at $\delta = 10^{-5}$, verifying Proposition 4.5 on the same nonsingleton slice.

The preceding corank-two line is diagonal: the relevant singular dyads are fixed. In the notation of Proposition 5.6, both $P_{\text{big}}(s)$ and $\tilde{B}(s)$ are constant. Hence $\partial_s \mathcal{W} = 0$, which proves $t_1 = 0$ and $\mathcal{M}_R = 0$ rather than merely observing their cancellation. To test a nonzero first-order coefficient without random data, take

$$A^* = \text{diag}(1, 3/5, 0, 0), \quad \lambda^* = 1,$$

$$\Theta = \begin{pmatrix} 1/6 & 0 & 2/5 & 0 \\ 0 & 1/6 & 0 & 3/10 \\ 1/5 & 0 & 2/3 & 0 \\ 0 & 1/10 & 0 & 1/3 \end{pmatrix}, \quad G = A^* - \Theta.$$

In the displayed coordinates, $a = 1/3$, $C = \text{diag}(2/3, 1/3)$, $\beta = 1$, and $\eta_R = 0.644812526657$. At $\delta = 10^{-5}$, the multiplier ratio is

$$(\lambda_\delta - \lambda^*)/\delta = -0.644805516183.$$

The direction ratio is 0.671768177074, already exhibiting a nonzero linear direction term. Direct evaluation of (63)–(64) gives

$$t_1 = 0.7010530600, \quad \|\mathcal{M}_R\|_F = 0.6717753807.$$

At $\delta = 10^{-6}$, the observed direction ratio is 0.671774660292, and the cosine between $(\Phi_\delta - \Phi_R^*)/\delta$ and $\mathcal{M}_R$ equals one to ten displayed digits (its computed value is 0.999999999997). The finite-scale estimate of t_1 at $\delta = 10^{-5}$ is 0.701047395444. Thus the table verifies the predicted vector coefficient, not only its exponent. Writing $t_\delta = (\lambda_\delta - 1)/\delta$, the complete sequence is

δ	t_δ	$\ \Phi_\delta - \Phi_R^*\ _F/\delta$	$(t_\delta - t_0)/\delta$
10^{-2}	−0.637857238093	0.664828202840	0.695528856469
10^{-3}	−0.644112038622	0.671057559126	0.700488035110
10^{-4}	−0.644742427015	0.671703367737	0.700996427320
10^{-5}	−0.644805516183	0.671768177074	0.701047395444
10^{-6}	−0.644811825605	0.671774660292	0.701052493549

This demonstrates sharpness for one fixed instance; no generic measure statement is inferred.

The compact-root branch of Corollary 5.7 is sharp as well. For $A^* = \text{diag}(2, 1, 0)$, $\Theta = \text{diag}(1, -1, 1)$, and $G = A^* - \Theta$, one has $a = 0$, $C = (1)$, $J_* = \text{diag}(1/4, 1, 0)$, and $\chi_* = -3/4$. Thus the predicted coefficients are $-3/8$ and $\|\mathcal{N}_0\|_F = \sqrt{26}/8 = 0.6373774392\dots$. At $\delta = 10^{-4}$, high-precision bisection gives the normalized pair $(-0.3749999965, 0.6373774341)$; the assertion script checks monotone convergence over four scales. Thus, even when the compact-polar equation already has a root, the nearby smooth path has nonzero errors beginning at cubic and quadratic order.

The doubly cancelled branch in (70) uses

$$A^* = \text{diag}(3, 2, 1, 0), \quad \lambda^* = 1,$$

$$\Theta = \begin{pmatrix} 27/10 & 0 & 0 & 3/10 \\ 0 & -16/5 & 0 & 1/10 \\ 0 & 0 & 1/2 & 1/4 \\ 1/5 & -3/20 & 1/10 & 1 \end{pmatrix}, \quad G = A^* - \Theta.$$

Here $a = 0$, $C = (1)$, $d_0 = 1$, $\chi_* = 0$, and $\chi_2 = 1/3$. Hence the predicted quintic coefficient is $-1/8$, and the quartic direction coefficient after subtracting $-\delta^2 J_*/2$ is

$$\text{diag}\left(\frac{1}{216}, \frac{3}{128}, \frac{3}{8}, -\frac{1}{8}\right).$$

Extended-precision bisection gives

δ	$(\lambda_\delta - 1)/\delta^5$
10^{-2}	−0.1248207361
10^{-3}	−0.1249833498
10^{-4}	−0.1249983476
10^{-5}	−0.1249998349

At $\delta = 10^{-4}$, the diagonal of the normalized quartic direction is

$$(0.004629629625, 0.023437499951, 0.374999996875, -0.124998347562),$$

while the Frobenius norm of its off-diagonal part is below 4×10^{-6} . The high-precision assertion script checks all four scales and the full matrix coefficient.

For Theorem 5.16 we use the one-parameter family $b_1 = b_2 = 1$, $c = 1 + \varepsilon$, for which $\beta_0 = \kappa_0 = \frac{1}{2}$ and $\Delta_\varepsilon = (c^2 - 1)/(1 + c^2)$. Choosing $\varepsilon = \varepsilon(\delta)$ so that $\Delta_\delta = \zeta \delta^{2/3}$ places the family in the critical window (92), whose limit solves $\frac{1}{2}y^3 + \zeta y^2 = 1$.

ζ	δ	$x_\delta/\delta^{2/3}$	y_ζ
0	10^{-9}	1.259920256195	1.259921049895
1	10^{-9}	0.839284832515	0.839286755214
3	10^{-9}	0.552471210127	0.552474610531
10	10^{-9}	0.313769660579	0.313775961214

For a valid strict-side moving path, take $\Delta_\delta = \delta^{1/3}$; then $\Delta_\delta \rightarrow 0$ while $\Delta_\delta/\delta^{2/3} \rightarrow \infty$. The normalization in (93), whose limit is one for this family, gives

δ	$x_\delta \sqrt{\Delta_\delta}/\delta$	$x_\delta/\widehat{x}_\delta$
10^{-6}	0.979950781607	0.985083206619
10^{-9}	0.997994100405	0.998500862499
10^{-12}	0.999799770023	0.999850008712
10^{-15}	0.999979992294	0.999985000087

This path respects the moving-family quantifier. By contrast, fixing ε and sending δ to zero tests the fixed strict-kink law, not the crossover theorem.

Finally, the oracle returns a matrix direction, not only a multiplier. At the boundary instance $b_1 = b_2 = c = 1$, direct differentiation gives

$$\|\mathcal{M}_0\|_F = \sqrt{3/2}, \quad \frac{\|\Phi_\delta - \Phi^*\|_F}{x_\delta} \rightarrow \sqrt{3/2}.$$

At $\delta = 10^{-15}$, the computed ratio is 1.224744871327, versus $\sqrt{3/2} = 1.224744871392$. The same verification includes critical, strict-side, boundary, and deliberately oscillatory paths for the unified ratio $x_\delta/\widehat{x}_\delta$.

For completeness, the oscillatory sequence is

$$q_k = 10^{-k}, \quad \delta_k = q_k^3, \quad \Delta_k = \begin{cases} 0, & k \equiv 0 \pmod{3}, \\ 3q_k^2, & k \equiv 1 \pmod{3}, \\ q_k, & k \equiv 2 \pmod{3}. \end{cases}$$

It alternates among boundary, critical, and strict-side windows, so the unified ratio has a limit even though the regime changes along the sequence. The displayed ones are rounded values, not exact identities.

k	Window	δ_k	$x_{\delta_k}/\widehat{x}_{\delta_k}$
2	strict	10^{-6}	0.985083206619
3	boundary	10^{-9}	0.999999370040
4	critical	10^{-12}	0.999999952844
5	strict	10^{-15}	0.999985000087
6	boundary	10^{-18}	0.999999999999
7	critical	10^{-21}	1.000000000000
8	strict	10^{-24}	0.999999985000
9	boundary	10^{-27}	1.000000000000
10	critical	10^{-30}	1.000000000000

The separate `verify_certificate_example.py` illustrates why a certificate must be assessed on the actual returned matrix. For $G = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$ and $\Theta = e_1 e_1^\top$, the exact completed direction is $\begin{pmatrix} 0 & 1/2 \\ 1/2 & 3/4 \end{pmatrix}$. It is tangent and optimal, whereas the compact polar factor at the same multiplier has tangency residual $1/5$. The script also compares an eight-step float64 polynomial map using the published Newton–Schulz coefficients from Xie et al. [2]. That diagnostic is not the production BF16 implementation, a Fenchel-path assertion, or a GPU-training experiment; its floating-point gap is not an interval-certified bound.

We emphasize what these tables do and do not support. They confirm the exponents and leading constants of the local laws on instances where those constants are known in closed form. They are not evidence about how often rank-deficient crossings occur in any application, and no such claim is made.

References

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- [3] Greg Yang, James B. Simon, and Jeremy Bernstein. A spectral condition for feature learning, 2023. Revised May 14, 2024.